Siqi Yan

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PROFESSIONAL SUMMARY

10+ years of experience in C/C++, Python, MATLAB, Linux and ROS development. Skilled in robotic vision/SLAM/navigation and embedded system design. Interdisciplinary experience in academic research, algorithm design, PCB soldering, firmware implementation and system deployment following first principles. Possesses full-stack development capabilities from low-level hardware to high-level algorithms.

TECHNICAL SKILLS

- Programming & Frameworks: C/C++, Python, MATLAB, Linux, ROS, g2o, RTAB-Map, PyTorch, OpenCV
- Robotics Platforms: NVIDIA Jetson, Arduino, Raspberry Pi
- Tools & Validation: PCB Soldering, Firmware Development, Simulation-to-Physical Validation

EDUCATION

University of Alberta, Edmonton, Canada | Studies toward PhD in Transportation Engineering Jan 2022 - Aug 2025

- Research focused on pose graph optimization for mapping in dynamic road environments. Designed and implemented graph-based nonlinear optimization pipeline using g2o framework.

University of Alberta, Edmonton, Canada | MSc in Computing Science Sep 2019 - Jan 2022

- Thesis: Improving Automated Vehicle Safety with Collective Perception.
- Research focused on multi-view graph-based optimization for simulated scene reconstruction.

University of Alberta, Edmonton, Canada | BSc in Computing Science Sep 2013 - Dec 2018

PROFESSIONAL EXPERIENCE

University of Alberta, Edmonton, Canada | Graduate Research Assistant Jan 2022 - Aug 2025

- Led a research theme focused on developing vision-based detection, tracking and wireless communication systems for Connected Intelligent Transportation Systems (C-ITS) use cases.
- Conducted field tests and system demonstrations, validating algorithms and system performance in real-world environments.
- Collaborated with a commercial software team on mobile application development.
- Developed a large-scale road construction zone mapping system using RTAB-Map and stereo cameras.
- Collected and analyzed data from 50 total round trip with 20 different drivers on Alberta highways.
- Led the design and development of a Python-based data analysis application, enabling non-technical team members to independently process and analyze traffic data, and increasing data analysis efficiency by 200%.

Maparobo Inc., Calgary, Canada | Software Developer Jun 2023 - Feb 2025

- Contributed to the development of mobile robotics applications using ROS, Python and C++.
- Designed a simulation-to-physical validation workflow to accelerate development despite limited hardware availability.

University of Alberta, Edmonton, Canada | Graduate Teaching Assistant Sep 2019 - Jan 2022

- Taught CMPUT 412 Experimental Mobile Robotics and CMPUT 379 Operating Systems.
- Graded assignments, held office hours, and guided students on course projects.

ComplexSys Solution, Edmonton, Canada | Software Developer Jan 2019 - Apr 2019

- Contributed to the development of a mobile robot navigation application for hotel service
- Developed terminal tools for testing and validating the robot behavior

University of Alberta, Edmonton, Canada | Graduate Research Assistant Feb 2018 – Apr 2018

- Worked on improving MoHex3HNN (an AlphaGo-Zero liked program for playing Computer Hex)
- Participated in the 2018 Computer Olympiad, and our program won the tournament

RELEVANT EXPERIENCE

Open Source Project: BananaKit2 Development Platform Jan 2025 - Present

- Designed and built a modular embedded systems development platform. Implemented the platform firmware using C and C++. Project link: https://github.com/siqiyan/BananaKit2

University of Alberta RoboMaster | CS Team Developer Jan 2017 - Dec 2019

- Designed and assembled multiple fully/semi-automated robots and drones.
- Participated as a core team member in the ICRA 2018/2019 RoboMaster AI Challenge, receiving finalist and 3rd prize.

PUBLICATIONS & AWARDS

- Haibo Cui, et al, Vision-Based Work Zone Safety Alert System in a Connected Vehicle Environment, Transportation Research Record Journal 2023.
- Chao Gao, Siqi Yan, Ryan Hayward, and Martin Muller, A Transferable Neural Network for Hex, ICGA Journal 2018.
- IEEE International Conference on Robotics and Automation 2019 RoboMaster AI Challenge, 3rd Prize (Montreal, Canada)
- RoboMaster 2018 International Regional Competition 3rd Prize (Shenzhen, China)
- IEEE International Conference on Robotics and Automation 2018 DJI RoboMaster AI Challenge Finalist (Brisbane, Australia)